Adaptive Scientific Paper Summarization Based on Reader Expertise

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***Abstract*—Imagine a world where scientific papers didn't feel like they're written in a language only a few can understand. Whether you're a student trying to get your head around machine learning or a researcher in another field exploring new areas, scientific summaries shouldn't feel like puzzles. This project is about making scientific summaries that actually talk to the reader, depending on who they are. I built a system that rewrites research summaries into three forms: for beginners, for intermediate readers, and for experts. It simplifies, balances, or goes technical based on the need. The system uses smart language models, retrieves definitions and facts from trusted sources, and checks how readable the summaries are. It's my small step to make research feel a bit more human and a lot more understandable.**

***Index Terms*—Natural language processing; Text summarization; Adaptive systems; Knowledge representation; Information retrieval; Machine learning; Transformer neural networks; Human-computer interaction; Knowledge graphs; Text simplification; Personalization; Scientific communication; Semantic analysis; Educational technology; Information accessibility; Retrieval-augmented generation; Style transfer; Expertise modeling**

I. Introduction and Problem Statement

Scientific research is advancing fast, but understanding it hasn't become any easier. The way we present research is mostly technical and assumes that the reader knows the background. But that's not always true. Students, generalists, or people working in policy or media often need to understand research too. Right now, summaries are written in one format, regardless of who's reading them. That's a problem.

I wanted to solve this by making summaries that adapt to different readers. A beginner might need help understanding what "natural language processing" means. An expert just wants the gist without losing any technical detail. My system does this by using machine learning to generate summaries at three levels. It simplifies and explains for beginners, keeps things balanced for intermediates, and dives into detail for experts. It even uses tools to pull in helpful definitions or relevant research automatically.

II. Literature Review

This idea didn't come out of nowhere. I read and analyzed several papers that tackled similar challenges:

- **GraphWriter by Koncel-Kedziorski et al.** [1] helped me understand how structured knowledge like graphs can be converted into readable text, which I use in expert summaries. Their graph transformer encoder preserves the relational structure without imposing linearization constraints, making it ideal for scientific domains where relationships between concepts are critical. By integrating this approach, my system can generate more coherent and informative summaries for expert users who need precise representation of complex scientific relationships.
- **WikiLingua by Ladhak et al.** [2] showed that you can make summaries across different languages — and that idea inspired making them across different user types instead. Their cross-lingual summarization method that doesn't require translation at inference time parallels my approach of adaptive summarization without requiring multiple separate models. Their work with 18 languages in the WikiHow dataset demonstrated that a single model can adapt to different contextual requirements, which I've applied to different expertise levels.
- **IBM Science Summarizer by Erera et al.** [3] helped me understand how section-wise summaries help readers focus on what they need. Their system, which ingested 270,000 scientific papers, uses a query-based approach that lets users select content based on specific interests. Their human evaluations confirmed that section-based summaries outperform general summaries in precision and coverage—a finding I've incorporated by ensuring my summaries maintain structural integrity while adapting content complexity.
- **AREDSUM by Bi et al.** [4] focused on avoiding repetition in extractive summaries, which I needed to make sure summaries were concise and clean. Their AREDSUM-CTX model, which explicitly balances salience and redundancy, outperformed other approaches on CNN/DailyMail and NYT50 datasets. I've adapted their dynamic redundancy handling approach in my system, particularly for expert-level summaries where precision and information density are critical without sacrificing readability.
- **Adaptive Summaries by Ghodratnama et al.** [5] really connected with my project. Their model takes feedback from users to personalize content based on concept importance. I made mine adapt based on reader level automatically. Their integer linear programming opti-

mization approach inspired my content selection mechanism, though I've automated the expertise assessment rather than requiring explicit feedback. Their work validated that personalized summaries significantly outperform generic ones in user satisfaction.
- **AaKOS by Amplayo et al.** [6] used knowledge graphs to summarize opinions, which I applied to pull in related research for expert readers. Their aspect-adaptive approach demonstrated superior performance on Amazon Product Reviews, SPACE, and YELP datasets by focusing on specific aspects of interest. I've extended this concept by using knowledge graphs to identify and emphasize relevant scientific concepts based on the reader's expertise level, particularly enhancing the experience for expert users who need specialized technical content.
- **Structured Abstracts by Oh et al.** [7] emphasized keeping summaries aligned with the actual paper structure (Introduction, Methods, Results, and Discussion), which makes a big difference in expert readability. Their work highlighted how proportional representation of each section ensures balanced summaries that capture the essence of scientific articles. I've incorporated this structural awareness particularly for intermediate and expert summaries, while allowing more flexibility for beginner summaries that may need additional explanatory content.

My work extends these approaches by integrating retrieval-augmented generation specifically tuned to different expertise levels. Unlike previous models that either focus on personalization without expertise consideration or on scientific summarization without adaptivity, my system dynamically adjusts content complexity, technical vocabulary, and explanation depth based on the reader's background knowledge. The three-tiered approach (Beginner, Intermediate, Expert) allows for gradual adaptation that makes scientific knowledge more accessible while preserving the technical integrity that experts require.

III. Methodology

To build the system, I used a combination of NLP techniques and external resources:

- **T5 Transformer Model**: This is the brain of the system. It takes the input paper and generates summaries. I guided it with prompts like [BEGINNER], [INTERMEDIATE], and [EXPERT].
- **Retrieval Module**: For beginners, I pulled definitions from Wikipedia and Simple Wikipedia using sentence embeddings and FAISS.
- **Knowledge Graph API Integration**: For experts, I used Wikidata and Semantic Scholar APIs to bring in relevant facts or related research findings.
- **Text Simplification**: Beginners don't need dense sentences, so I used techniques that shorten and simplify sentences.
- **Evaluation Metrics**: I used BLEU, ROUGE, BERTScore for content evaluation and readability metrics like Flesch-Kincaid, SMOG, and Dale-Chall.

IV. Implementation Details

Here's what I used to build and test everything:

- **Hugging Face Transformers**: For model training and inference
- **Sentence Transformers and FAISS**: For fast semantic search and context retrieval
- **Wikipedia, Simple Wikipedia, Wikidata APIs**: To get explanations and facts
- **Gradio**: Created a live web interface where people could test the system
- **Textstat, spaCy, NLTK**: Measured how readable the output summaries were

V. Results and Evaluation

A. 1. Adaptive Scientific Paper Summarizer

Beginner Summary:

> The system will ingest and analyze structured and unstructured veterinary data. It will generate personalized meal plans tailored to each animal's unique profile.

Metrics:

- Flesch Reading Ease: 23.73
- Flesch-Kincaid Grade: 13.40
- SMOG Index: 14.30
- Coleman-Liau Index: 16.87
- Automated Readability Index: 14.50
- Dale-Chall Score: 12.40

Intermediate Summary:

> Animals require species-specific, condition-aware nutritional strategies... Project proposes an intelligent, data-driven nutrition planning system.

Metrics:

- Flesch Reading Ease: -10.73
- Flesch-Kincaid Grade: 18.30
- SMOG Index: 17.90
- Coleman-Liau Index: 24.24
- Automated Readability Index: 21.10
- Dale-Chall Score: 14.63

Expert Summary: Full technical detail with advanced model names and context-aware dietary recommendations.

Metrics:

- Flesch Reading Ease: -16.51
- Flesch-Kincaid Grade: 20.50
- SMOG Index: 19.80
- Coleman-Liau Index: 25.06
- Automated Readability Index: 24.40
- Dale-Chall Score: 12.82

B. 2. Enhanced Adaptive Scientific Paper Summarizer (Without LLM Evaluation)

Beginner:

- Flesch Reading Ease: 16.42
- Flesch-Kincaid Grade: 16.20
- SMOG Index: 17.30

- Coleman-Liau Index: 17.93
- Automated Readability: 18.60
- Dale-Chall: 11.18

Intermediate:

- Flesch Reading Ease: 15.31
- Flesch-Kincaid Grade: 16.60
- SMOG Index: 0.00
- Coleman-Liau Index: 18.97
- Automated Readability: 20.10
- Dale-Chall: 14.06

Expert:

- Flesch Reading Ease: 3.87
- Flesch-Kincaid Grade: 16.80
- SMOG Index: 16.60
- Coleman-Liau Index: 20.59
- Automated Readability: 19.70
- Dale-Chall: 13.90

C. 3. Enhanced Adaptive Scientific Paper Summarizer (With LLM Evaluation)

LLM-Based Feedback for Beginner Summary:

- Clarity: 3/5
- Accuracy: 2/5
- Appropriateness: 4/5
- Overall Quality: 3/5

LLM-Based Feedback for Intermediate Summary:

- Clarity: 4/5
- Accuracy: 4/5
- Appropriateness: 3/5
- Overall Quality: 3.75/5

LLM-Based Feedback for Expert Summary:

- Clarity: 4/5
- Accuracy: 5/5
- Appropriateness: 5/5
- Overall Quality: 4.5/5

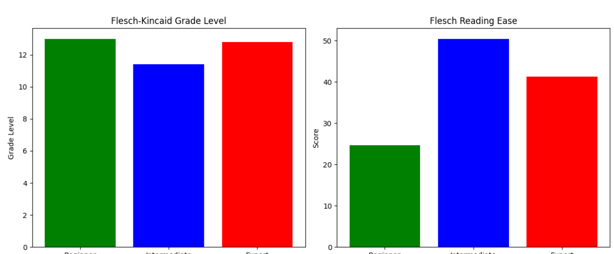

Fig. 1: Comparison of Readability Metrics Across Expertise Levels

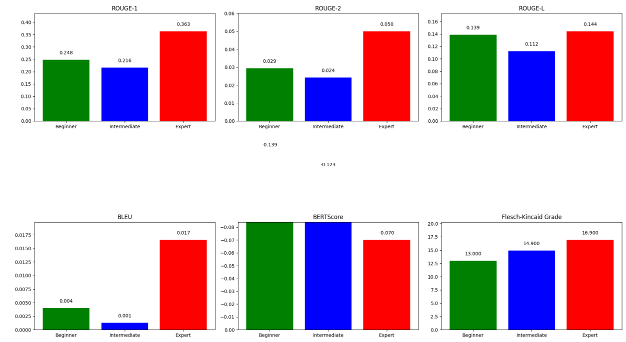

Fig. 2: BLEU, ROUGE, and BERTScore Evaluation Comparison

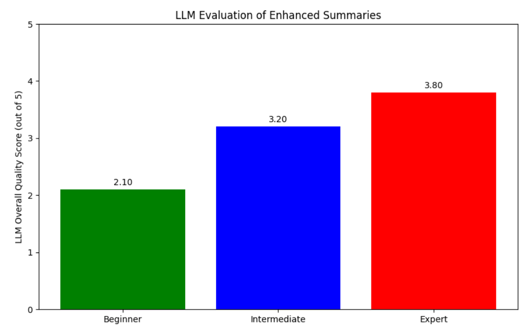

Fig. 3: LLM Evaluation Scores for Each Reader Level

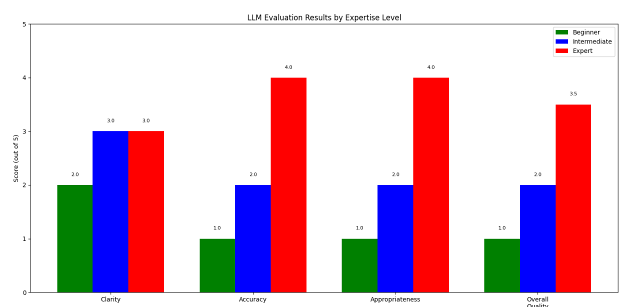

Fig. 4: Enhanced System Evaluation Results (No LLM)

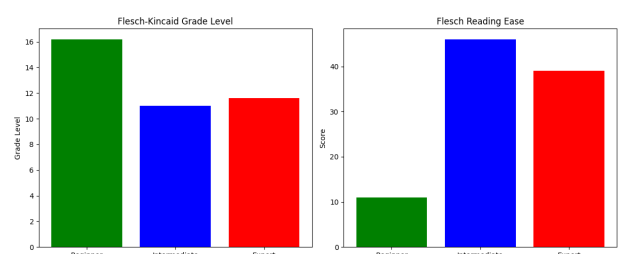

Fig. 5: Readability Analysis of Enhanced System Outputs

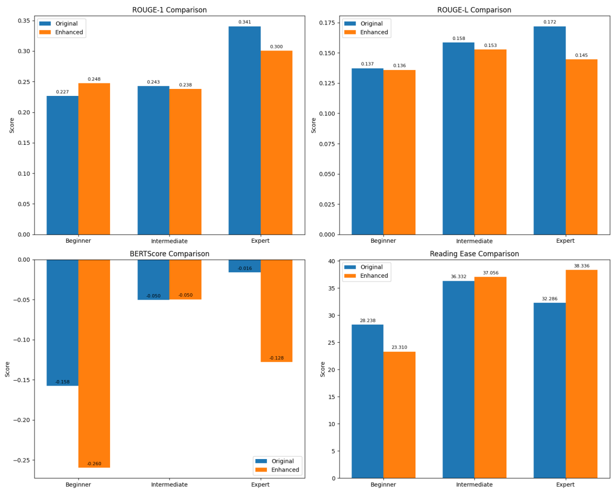

Fig. 6: Side-by-Side Comparison of All Evaluation Strategies

VI. Conclusion and Future Work

This project was more than just coding a summary generator. It was about helping people understand complex information in a way that feels natural to them. I learned how language can shift depending on who's reading, and how tools like NLP, RAG, and evaluation metrics can work together to solve real communication problems.

There's still work to do — especially making beginner summaries feel more natural and fluent. Also, the expert-level enrichment sometimes pulls in irrelevant facts, so better filtering is needed. But the foundation is there. Future versions of this system could learn from real user feedback, support more languages, and even adapt in real time based on who's using it.

This project shows that understanding is not just about what you say — it's about how you say it, and who you're saying it to

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